

Subject: Submission of a new gridded multi-sector water-demand dataset for CONUS, 1980–2100

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Dear Editor,

We are pleased to submit our manuscript, “**High-resolution monthly sectoral water demands for the U.S. over 1980–2100**”, for consideration in *Scientific Data*. The accompanying dataset is publicly archived on MSD-Live with a citable DOI, and the analysis code is openly available on GitHub.

We present a gridded ($1/8^\circ$), monthly, multi-sector water-demand dataset for the contiguous United States covering 1980–2100 across one historical period and eight future scenarios that span a wide but plausible range of atmospheric, emissions, economic, technological, and population conditions. The dataset resolves six demand sectors (irrigation, thermoelectric, municipal, livestock, manufacturing, and mining), aggregated from 25 underlying GCAM-USA subsectors, separately for withdrawals and consumption, and includes per-cell groundwater/surface-water source attributions.

The dataset provides regional advances to the most directly comparable global product (Khan et al. 2023): (1) state-resolved sectoral demands from GCAM-USA, (2) explicit Capacity Expansion Regional Feasibility (CERF) power-plant siting consistent with capacity-expansion projections through 2100, (3) SSP-consistent population proxies, (4) growing-season-index-based temporal downscaling for irrigation, (5) USGS-anchored groundwater/surface-water source attribution, and (6) a $4\times$ spatial-resolution refinement ($16\times$ cells).

We provide detailed comparison of the historical record against the USGS 2010–2020 water-use reanalysis at HUC6 for the three largest sectors, with Pearson correlations between 0.73 and 0.95. We expect the dataset to support hydrologic, energy-water-land, and integrated multisector dynamics modeling, as well as research on bulk-power-system resilience, water scarcity, and groundwater/surface-water interactions.

We would be glad to provide a list of suggested reviewers with relevant expertise in integrated water-energy-land modeling, water-demand downscaling, and water-resources data products on request. A preprint of the manuscript is being deposited on EarthArXiv in parallel with this submission.

Sincerely,

Cameron Bracken (corresponding author), Pacific Northwest National Laboratory, cameron.bracken@pnnl.gov, ORCID: 0000-0003-1917-402X

Hassan Niazi (corresponding author), Joint Global Change Research Institute, Pacific Northwest National Laboratory, hassan.niazi@pnnl.gov

on behalf of all co-authors: Travis Thurber, Isaac Thompson, Kazi Tamaddun, Hisham Eldardiry, Kendall Mongird, Nathalie Voisin, Ning Sun, Jennie Rice